

PROCESS MANUAL – AMINE TREATING UNIT (OPERATING ENVELOPE)

RESTRICTED – process chemistry. Distribution limited to Operations, Process Engineering and Technical Services.

1. Operating envelope

Parameter	Normal	Limit	Consequence of deviation
Lean amine strength (MDEA)	45–50 wt%	40–52 wt%	Low: poor H ₂ S removal; high: viscosity, foaming
Rich amine loading	0.35–0.42	≤ 0.45 mol H ₂ S/mol amine	Corrosion in rich lines, E-501
Lean amine loading	< 0.01	0.02 mol/mol	Treated gas off-spec
Antifoam dosing	5–10 ppm	max 20 ppm	Overdosing promotes foaming / fouling
Regenerator V-501 bottom temp	121–124 degC	127 degC	Amine degradation
Lean amine temp to absorber	5 degC above gas	-	Hydrocarbon condensation / foaming

2. LPG caustic wash

- Caustic (NaOH) strength for LPG treating: 5–10 wt% NaOH.
- Replace caustic when spent > 70% (by titration).
- Water wash downstream to remove entrained caustic.

3. Foaming response

- Indications: erratic absorber dP, level swings in V-502, amine carry-over.
- Action: inject antifoam 5–10 ppm, check carbon filter, reduce throughput; sample for hydrocarbons.